

AksharaScan

Everything in this project, explained in plain words — what it does, why it matters, what the numbers mean, and what to say when a judge asks you a hard question.

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You do not need to memorise this. Read it once, slowly. Then read only two sections again before you stand at the table: **section 9** (what to say in the first thirty seconds) and **section 11** (the questions judges ask). Everything else is here so that nothing surprises you.

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1 What you built, in one breath

You built a way for someone who can move only one muscle to write Malayalam — and you measured, with real numbers, how much faster it is than what such a person has to use today.

It is three things together:

An app

Runs on a tablet. Shows Malayalam letters and moves a highlight through them. One button selects.

A switch box

Two big buttons wired to a small circuit board that talks to the tablet over Bluetooth, pretending to be a keyboard.

An experiment

A measurement of how many button presses each method costs, run on 113,324 units of real Malayalam.

And a finding

Including one that proved your own starting idea wrong — which, done properly, is the most scientific part of the whole project.

2 Who this is for

Think of a person with cerebral palsy, or motor neurone disease, or a spinal cord injury high up the neck. Their mind is completely intact. They know exactly what they want to say. But they cannot speak clearly, and they cannot use a keyboard or a touchscreen, because they cannot reliably move their hands.

What they often *do* have is **one reliable movement**. A tap of one finger. A press of the head against a cushion switch. A puff of air. A twitch of one cheek muscle that a sensor can see.

That single movement is their entire connection to the world. Everything they will ever write has to come through it.

THE PERSON EVERYONE HAS HEARD OF

Stephen Hawking typed exactly this way. After he lost his voice in 1985, a cursor moved across rows and letters on his screen and he squeezed a clicker in his hand to choose. At his best he managed **about 15 words a minute**.

Then his hands failed too. The clicker was replaced by an infrared sensor mounted on his glasses, watching a small twitch in his cheek. **Same brain, same software, worse switch** — and his speed collapsed to **one or two words a minute**.

Use this story. It explains in fifteen seconds why the number of presses matters so much, and a judge will instantly understand it.

And here is the gap your project is about. All of this work — decades of it — was done for English. A Malayalam speaker gets the same tools with the Malayalam letters simply dropped in, with none of the optimisation. **So they press more times to say the same thing.**

3 How typing with one button works

This is the single most important thing to be able to explain, and the one thing nobody guesses on their own. Practise saying it out loud.

SAY THIS, AND HAND THEM THE SWITCH

“The screen asks you a yes-or-no question. **Press means yes. Waiting means no.** That is the whole thing. Try it — type your own name.”

Why does waiting mean no? Because the person has only one button. They cannot press a separate “no” button. So the app uses *time itself* as the second answer: it shows you a group of letters for two seconds, and if you do nothing, it takes that as “not this group” and moves on.

That is why the project counts two different things, and you must keep them apart in your head:

WHAT IT IS

WHY IT MATTERS

Presses

how many times the button is pushed

This is the *effort*. For someone who tires quickly, or whose movement sometimes misfires, every press is expensive and risky.

Scan steps

how many groups the highlight moves through

This is the *time*. Each step takes about two seconds, so steps are literally seconds of someone's life spent waiting.

A good system reduces both. A careless comparison reduces one while quietly making the other worse — and that actually happened in this project, which is in section 8.

The twenty questions idea

Suppose you want one letter out of 50. The slow way is to show all 50 one at a time and wait for the press — on average 25 steps.

The fast way is the game of twenty questions. Split the letters into two halves. Ask: “is your letter in this half?” Press means yes, wait means no. Either way you have **thrown away half the alphabet in one answer**. Do it again, and again — and about six answers get you to one letter out of 50.

That is what the project calls a **scan tree**: a pre-planned sequence of yes-or-no questions that gets to any letter in as few answers as possible.

4 Why Malayalam is harder than English

English has 26 letters in a flat row. You arrange them so that common letters need fewer questions, and you are finished. This was solved decades ago.

Malayalam is not a flat row. It is an **akshara** script, and it has structure that English simply does not have.

One written shape is often several pieces

Take ഏ. That is a single written shape, but it is built from three separate things: the consonant ഐ, a silencing mark called a *virama* (്), and then ഏ. A combination like this is a **koottaksharam** — a conjunct.

So a single thing the reader sees as “one letter” can cost three separate selections. English never does this.

Vowels hang off consonants

In English, *k* and *a* are two letters side by side. In Malayalam, ഐ plus the vowel sign for *aa* becomes ഐ — the vowel is a mark attached to the consonant, not a letter standing beside it. Some vowel signs even appear to the *left* of the consonant they belong to, though they are typed after it.

And there are rules about what can follow what

You cannot put a vowel sign straight after another vowel sign. You cannot put a vowel sign immediately after a virama. These are not style preferences — they produce text that is simply not Malayalam.

THIS IS THE NUMBER THAT IS GENUINELY YOURS

Because of all this, the same sentence costs **2.3 times more scan steps** in Malayalam than the equivalent effort in a flat alphabet. The project calls this the **akshara tax**.

Nobody has ever published that figure for any Indian script. When a judge asks what is new here, this is one of your two best answers. It is a real measurement of a real disadvantage that millions of people pay and nobody had counted.

5 The clever part: asking good questions

Twenty questions is only fast if you ask *good* questions. Two ideas make the questions good, and both are easy to explain.

Idea one: common letters should need fewer questions

Do not split the alphabet into two equal *halves by count*. Split it into two halves of equal *likelihood*. Put the letters people actually use a lot where they can be reached in two or three answers, and push the rare ones deeper.

This is the same mathematics that makes a ZIP file smaller: give short codes to common things and long codes to rare things. It is called **Huffman coding**, from 1952. You did not invent it, and you should say so.

Idea two: the questions should change depending on what came before

After you type ഓ, what comes next is not random. Some things are very likely, some are impossible. So the app **rebuilds the entire question tree after every single selection**, using what it learned from real Malayalam about what usually follows what.

Where did it learn that? From **113,324 units of real Malayalam** taken from Malayalam Wikipedia — about 11,170 words. The app is not using your guesses about the language. It counted.

Idea three, and this is the big one: guess the whole word

Why ask about letters at all, if you can offer whole words? Once a couple of letters are in, the app offers complete words and likely next words. Choosing a whole word from a list costs two or three presses instead of twenty.

This turns out to be the biggest win in the entire project — bigger than anything to do with the clever tree. More on that in section 8, because it is also the thing you must be careful not to take credit for.

6 What is actually in the box

The app

- **Three layers:** Malayalam letters, numbers (123), and English letters (ABC). All reachable using only the switch — no touching the screen.
- **Word prediction** that learns from use.
- **Speech.** It speaks the finished sentence aloud using the tablet's own Malayalam voice, with no internet needed. Five online voice services can be plugged in as an upgrade, and anything they produce is saved so it still works offline afterwards.
- **Quick phrases** — his own common sentences, about three presses each instead of four minutes.
- **A tutorial** that walks a new user through pressing, waiting, one letter, their name, their age, and sending it.
- **It works with no internet at all.** This matters: fair venues usually have no usable WiFi, and a demo that needs a network scores nothing.

The hardware

An **ESP32** — a small, cheap circuit board with Bluetooth built in. Two large buttons are wired to it. The board pretends to be a **Bluetooth keyboard**, so when a button is pressed the tablet simply receives a keypress.

WHY PRETENDING TO BE A KEYBOARD IS A GOOD DESIGN

Because the switches then work in *every* app on the tablet, not just yours. That is exactly how real commercial switch interfaces for disabled users behave — devices like the Blue2 and the Hook+. You did not do something odd; you did the standard thing, deliberately.

If a judge asks “could a real user use their own switch?” the answer is yes — add a 3.5 mm socket, which is the standard connector in this field, and their existing switch plugs straight in.

Two buttons, three commands: button one selects, button two deletes, and holding button two (or pressing both together) speaks the sentence.

The board needs nothing but **5 volts of power** — a phone charger or a power bank. No laptop is involved in using it.

7 The numbers, and what they mean

This is the heart of the poster. The test is typing എന്റെ പേര് ആദിത്യ — “my name is Adithya” — which is 17 units of Malayalam.

METHOD	SCAN STEPS	PRESSES	TIME AT 2 S/STEP
Row-column what a Malayalam user gets today	178	34	5 min 56 s
AksharaScan, letter by letter	77	44	2 min 34 s
AksharaScan + word prediction	33	18	1 min 6 s

Five minutes and fifty-six seconds, down to one minute and six seconds. For three words. That is the sentence a judge should hear.

Now look at the middle row carefully, because it contains something awkward and you should understand it before a judge spots it. Letter-by-letter AksharaScan cut the *steps* from 178 to 77 — but it *raised the presses* from 34 to 44.

UNDERSTAND THIS, IT IS A REAL TRADE-OFF

A question tree is **press-expensive and fast**. Row-column is **press-cheap and slow**. You are trading button presses for time.

For someone who tires easily, more presses might genuinely be the worse deal — and the honest position is to say so. **Word prediction is what resolves it:** 18 presses against the old 34, and far fewer steps too. Fewer of both. But only with prediction switched on.

8 The honest part: you were wrong

Read this section twice. It is the part most likely to win you respect from a serious judge, and the part most likely to go badly if you have not thought about it.

What you set out to prove

The original idea was this. Malayalam has hard rules about what cannot follow what. So: build the question tree so that **illegal letters are removed entirely** at every step. Never even offer something that cannot happen. That should save a lot of presses.

It is a sensible idea. It sounds obviously right.

What you actually measured

THE RESULT

Removing the illegal letters saved **0.0%**. Nothing. Not a little — nothing at all.

And this was not one unlucky test. It held across **eight separate experiments**: on your hand-written rules and on rules measured from real Wikipedia text; with legal sets ranging from 32% to 63% of the alphabet; on clean typing and with simulated mistakes; on single pieces and on whole conjuncts. **Same answer every time.**

Why — and this is the interesting bit

Because the tree had already solved the problem by accident.

Remember idea one: rare things get pushed deep into the tree. An *illegal* letter is just an extremely rare thing — so the tree had already buried it so deep that the highlight practically never reached it. Deleting something the user was never going to arrive at saves nothing.

SAY IT LIKE THIS

“Huffman coding already gives improbable symbols very long codes. So forbidding them outright changes almost nothing — the scanner was never getting there anyway. **Removing an impossible option only helps if your model wrongly believed it was likely.**”

Why this is a good result and not a failure

A judge may try to make this sound like your project failed. It did not, and here is why — in the order you should say it.

- 1 **You predicted, you built, you measured, and you reported what you found.** That is science. Announcing only results that flatter you is the opposite of science. Most projects at a fair have never tested their own central claim hard enough to be able to find it wrong.

- 2 **You can explain *why*, mathematically.** A negative result with a mechanism is a contribution; a negative result with a shrug is not. You have the mechanism.
- 3 **You found out what actually matters instead.** The measurement ranked the effects honestly: word prediction removes 57% of the steps, using a question tree at all is the next biggest, a context model after that — and the idea you started with, about 0%.
- 4 **Three real findings survived, and they do not need the original idea to be true.** They are below.

What survived

FINDING	WHY IT COUNTS
The akshara tax is 2.3×	Malayalam costs 2.3 times the scan steps. Never published for any Brahmic script. A real, new measurement.
Break letters apart, do not treat them as whole blocks — 19% better	A design recommendation with a number attached, which anyone building a Malayalam AAC app can use tomorrow.
Legality is free	It needs no text corpus at all — just the grammar of the script, which a linguist can write down in an afternoon. For the roughly twenty Indian languages with no usable corpus, that makes it worth having even though it adds nothing where a corpus exists.
The negative result itself	With its reason. This stops the next person wasting a year on the same idea.

ONE MORE FINDING WORTH HAVING READY

When the rules were learned from Wikipedia instead of written by hand, something revealing happened. A combination that is genuinely rare in real Malayalam appeared **3** times. A combination that was pure *junk* — caused by Wikipedia's formatting splitting words in half — appeared **57** times.

The noise was 19 times more common than the fact. So no amount of “ignore anything rare” could separate them. Only a structural rule could.

The lesson: *“just learn the rules from data” sounds clean and is not. A constraint learned from a corpus is only as good as the tool that split the text into pieces — and when it is wrong, it looks exactly like being right.* That is a genuinely sophisticated point and it is yours.

9 The first thirty seconds at the table

Judges walk up with no context and little time. Do not start with scan trees or Malayalam orthography. Do this instead.

In order, every time

- ☐ **One sentence about the person.** “Some people can move only one muscle. This is how they write.”
- ☐ **Put the switch in their hand.** Do not explain first. Hand it over.
- ☐ **Say the one rule.** “It asks you yes or no. **Press for yes. Wait for no.**”
- ☐ **Let them type their own name.** They will be slow. That is the point — they are feeling the problem in their own thumb.
- ☐ **Now show the counter.** Point at the press count and the timer. Then switch on prediction and let them do it again.
- ☐ **Only now, the science.** The $2.3\times$ tax, the 178-to-33 comparison, and the negative result.

The reason this order works: once someone has felt how slow it is, every number you say afterwards lands on an experience instead of on nothing.

THE SINGLE LINE TO REHEARSE

“It asks you yes or no. **Press for yes. Wait for no.**”

Nobody guesses that waiting is an answer. Say it before anything else, and say it exactly the same way every time.

10 Words you will need to explain

Judges will use some of these at you, and you will need others to answer. Keep the plain-words version ready — never the dictionary version.

AAC

Augmentative and Alternative Communication. Any tool that helps someone communicate when speech does not work. Your project is an AAC device.

Scanning

Moving a highlight through choices so a person with one button can pick one. The method, not the project.

Switch

The one button. In this field it means any single input — a button, a puff of air, a blink sensor.

Scan tree

The planned sequence of yes-or-no questions. A good tree asks fewer questions to reach common letters.

Row–column scanning

The old method you compare against: highlight each row until the user presses, then each letter in that row. It is what Android's accessibility tools actually ship today, which is why it is the fair baseline.

Akshara

The natural writing unit of Malayalam and related scripts — roughly a consonant with its vowel attached. The thing a reader perceives as one letter.

Koottaksharam / conjunct

Two or more consonants fused into one shape, like ക്ക. One shape, several pieces to type.

Virama

The small mark (്) that removes a consonant's built-in vowel. It is what glues conjuncts together.

Huffman coding

A 1952 method for giving short codes to common things and long codes to rare things. The same idea that makes files compress. Borrowed, not invented here.

Corpus

A large pile of real text used to count what is common. Yours is Malayalam Wikipedia — 113,324 units.

Word prediction

Offering whole words once a couple of letters are typed. The biggest single effect measured in the project, and the oldest idea in it.

Bluetooth HID

The standard that lets a device present itself as a keyboard or mouse. HID means Human Interface Device. It is why your switches work in every app.

ESP32

The small cheap circuit board with built-in Bluetooth that the buttons are wired to.

Low-resource language

A language with little digital text available. Most Indian languages are low-resource, which is exactly why the “legality is free” finding matters.

11 Questions judges ask

Answer briefly first, then stop. Let them ask for more. A short confident answer beats a long nervous one.

What does your project do?

“It lets someone who can move only one muscle write Malayalam, and it does it in about a fifth of the time the current method takes.”

Who would actually use this?

People with cerebral palsy, motor neurone disease, advanced multiple sclerosis, or high spinal injuries — where the mind is fine and the body is not. Mention Stephen Hawking; it makes it concrete immediately.

Doesn't this already exist?

“One-button scanning exists and is decades old. It has been optimised for English for decades. For Malayalam, nobody has done that work — the letters were swapped in and the optimisation was not. I measured the penalty that leaves behind, and it is 2.3 times.”

Be very clear that you are not claiming to have invented scanning. You are claiming to have measured and reduced a disadvantage specific to Malayalam.

Why not use eye tracking, or a brain interface?

Cost and reliability. An eye tracker is expensive and needs calibration and decent lighting. Brain-computer interfaces are research equipment with poor accuracy and long training times. A switch is a few hundred rupees, works in any light, never needs calibration, and is what the overwhelming majority of real users actually have. **Making the cheap thing better helps more people than making the expensive thing exist.**

Why not just use voice recognition?

These users often cannot produce reliable speech — that is frequently the reason they need the device at all.

What is new here? What is your actual contribution?

Have exactly three answers, in this order:

1. The **akshara tax of $2.3\times$** — measured for the first time for any script of this family.
2. **Decompose, do not cluster — 19%**. A usable design recommendation with a number.
3. A **negative result with its mechanism**: hard legality constraints buy nothing, and here is the information-theoretic reason.

Did your hypothesis work?

Answer this one head-on. Do not soften it.

“No. I predicted that removing illegal letters from the tree would save presses. I measured it eight different ways and it saves zero. The reason is that Huffman coding had already pushed those letters so deep that the scanner never reached them — so deleting them changes nothing. I reported it because I had pre-registered that I would report it either way.”

A judge who hears that delivered calmly will trust every other number on your poster.

So isn't your project a failure?

“The hypothesis failed. The project measured it correctly and explained why, and it also produced two measurements nobody had made before. A result that stops the next person repeating my mistake is worth having.”

How do you know your numbers are right?

Three independent checks, and say all three:

- The model is **measured from 113,324 units of real Malayalam**, not guessed.
- There are **131 automated tests**, including one that forces the live press counter on the tablet to agree exactly with the offline calculation — so the number a judge sees cannot drift from the number in the write-up.
- The same conclusion appears in **three independently built models** with legal sets from 32% to 63%.

What are you comparing against? Isn't your baseline unfair?

“Row-column scanning, which is what Android's own accessibility tools ship today. It is what a Malayalam user would really be using, not a weak version I invented to beat.”

Where did the Malayalam data come from?

Malayalam Wikipedia, about 450 random articles, fetched through their official interface. It is licensed CC BY-SA 4.0 and credited in the project's notice file. Say openly that it is Wikipedia — and that cleaning it was harder than downloading it, which leads into the 57-versus-3 noise story if they are interested.

Which part is the biggest improvement?

Word prediction — 57% of the steps. And then immediately add: **“which is also the least novel part. Intel's biggest win for Stephen Hawking was prediction too.”**

Volunteering this is important. If a judge works it out and you look as though you were hiding it, you lose the room.

Could a real disabled person use this today?

Be precise and honest. It works, it is installed on a tablet, it speaks, and it runs with no internet. But **it has not been tested with any disabled user**. That is the obvious and necessary next step, it needs ethical approval, and it is what you would do next.

Never claim it has been user-tested. One follow-up question destroys that.

Does it work for other languages?

The method is for akshara scripts generally — Hindi, Tamil, Kannada, Telugu, Bengali are built the same way. The approach transfers; the measurements would need redoing per script. The project uses English as a deliberate control, precisely because English should show no akshara tax.

How much did it cost?

The ESP32 and two buttons are a few hundred rupees. The tablet is a normal tablet. The app is free and the code is public. Low cost is a genuine strength here: AAC devices routinely cost lakhs, and that is why most Indian users have none.

What was the hardest part?

A good honest answer is the data cleaning: discovering that corpus *noise* appeared 57 times while the real rare pattern appeared 3 times, so no frequency cutoff could separate them. Or the one where the app typed a letter by itself when nobody pressed anything — the scanner reached a choice with zero presses, which had to be restructured.

What would you do next?

- Test with actual users who rely on switches.
- Plot how much the legality rules help as the amount of available text shrinks — the prediction is that it matters most where there is least text, which is the real situation for Indian languages.
- Offer the scan trees to an existing Malayalam AAC app so real users get them.

Did you build this yourself?

Say what is true, clearly and without embarrassment: what you designed and decided, what you measured, and what is borrowed — Huffman coding from 1952, the Wikipedia text, the standard Bluetooth keyboard libraries, and prior research on scan trees. Section 15 lists it. Judges respect a clear boundary far more than a vague claim to have done everything.

12 Four traps to avoid

TRAP 1 — SAYING “2.3 TIMES FASTER”

The $2.3\times$ is **scan steps**, which is time. Letter by letter it actually needed **more presses** than the baseline: 44 against 34. Say “*2.3 times fewer scan steps*”, and if you want the clean headline use the prediction row: **178 steps and 34 presses down to 33 and 18.**

TRAP 2 — DEFENDING THE HYPOTHESIS

If a judge presses on the legality idea, do not argue for it. It measured zero, eight times. Agree, explain the mechanism, and move to what survived. A student who defends a disproved claim looks worse than one who never tested it.

TRAP 3 — IMPLYING A REAL USER HAS TRIED IT

Nobody with a disability has tested this. If asked, say so plainly and call it the next step. Any exaggeration here is both dishonest and easy to catch.

TRAP 4 — LETTING PREDICTION LOOK LIKE YOUR INVENTION

Prediction is the largest effect and the oldest idea. Volunteer that, early. Your contribution is the akshara measurement and the honest negative result.

13 What you should admit freely

Saying these before you are asked makes you look like a scientist rather than a salesperson. Every one of them is already written down in the project.

- The central hypothesis was **wrong**, and it is reported as wrong.
- The **largest effect belongs to someone else** — word prediction, known since the 1980s.
- **No disabled user has tested it.**
- Letter by letter, it **costs more presses** than the baseline. Only prediction fixes that.
- The akshara tax of $2.3\times$ comes from **simulation**, not from timing real people.
- The corpus is **Wikipedia**, which is not how people chat. Real conversation would give different letter frequencies.

14 If something breaks

PROBLEM	WHAT TO DO
Switch does nothing	Check the ESP32 has power (a light is on), then Bluetooth on the tablet. Fall back to the space bar or screen taps — the app works without the hardware.
It types on its own	A button is stuck closed. Unplug the board; demo with the keyboard instead.
No sound	Tablet volume, then Settings → Voice → Check this device's voices . The app types fine without speech — say so and carry on.
Scanning feels too fast or slow	Speed control is top-left on the compose screen, changeable mid-sentence. Slower is better for a first-time judge.
Numbers look wrong between judges	Press ↺ Restart , top right. It clears the sentence, the press count and the timer.
No WiFi at the venue	Does not matter. Everything essential runs offline by design.

BEFORE YOU LEAVE HOME

Charge the tablet. Bring the power bank, both cables, and spare buttons if you have them. Put the tablet in **aeroplane mode and rehearse the whole demo offline** — if anything depends on the network you want to find out at home, not at the table.

15 What is yours and what is borrowed

Have this boundary clear in your head. If a judge asks and you answer confidently, it reflects well on everything else.

YOURS

The measurement of the akshara tax (2.3×)
The decompose-vs-cluster comparison (19%)
The eight experiments and the negative result
The corpus-noise finding (57 against 3)
The app, the scan-tree code, the firmware
The decision to report a disproved hypothesis

BORROWED, AND CREDITED

Huffman coding — Huffman, 1952
Optimal scan trees — Higger and others, 2016
Context-conditional trees — Roark and others, 2013
Scan-step definitions — MacKenzie, 2012
Word prediction — standard in AAC since the 1980s
Malayalam text — Malayalam Wikipedia, CC BY-SA 4.0
Malayalam font — Noto, SIL Open Font Licence
Bluetooth keyboard library, app wrapper — open source

THE LAST THING TO REMEMBER

You are not trying to sound like you invented everything. You are trying to sound like someone who **asked a precise question, built the apparatus to answer it, measured honestly, and knows exactly which parts of the answer are new.**

That is what a good judge is actually looking for, and it is a much easier thing to be than an inventor of everything.